

Ch-4 (NLM)

1st Law: $\rightarrow$

2nd Law

$$\text{Momentum}(\vec{p}) = m \times \vec{v}$$

$\uparrow$ vector
Dirn of $\vec{p}$

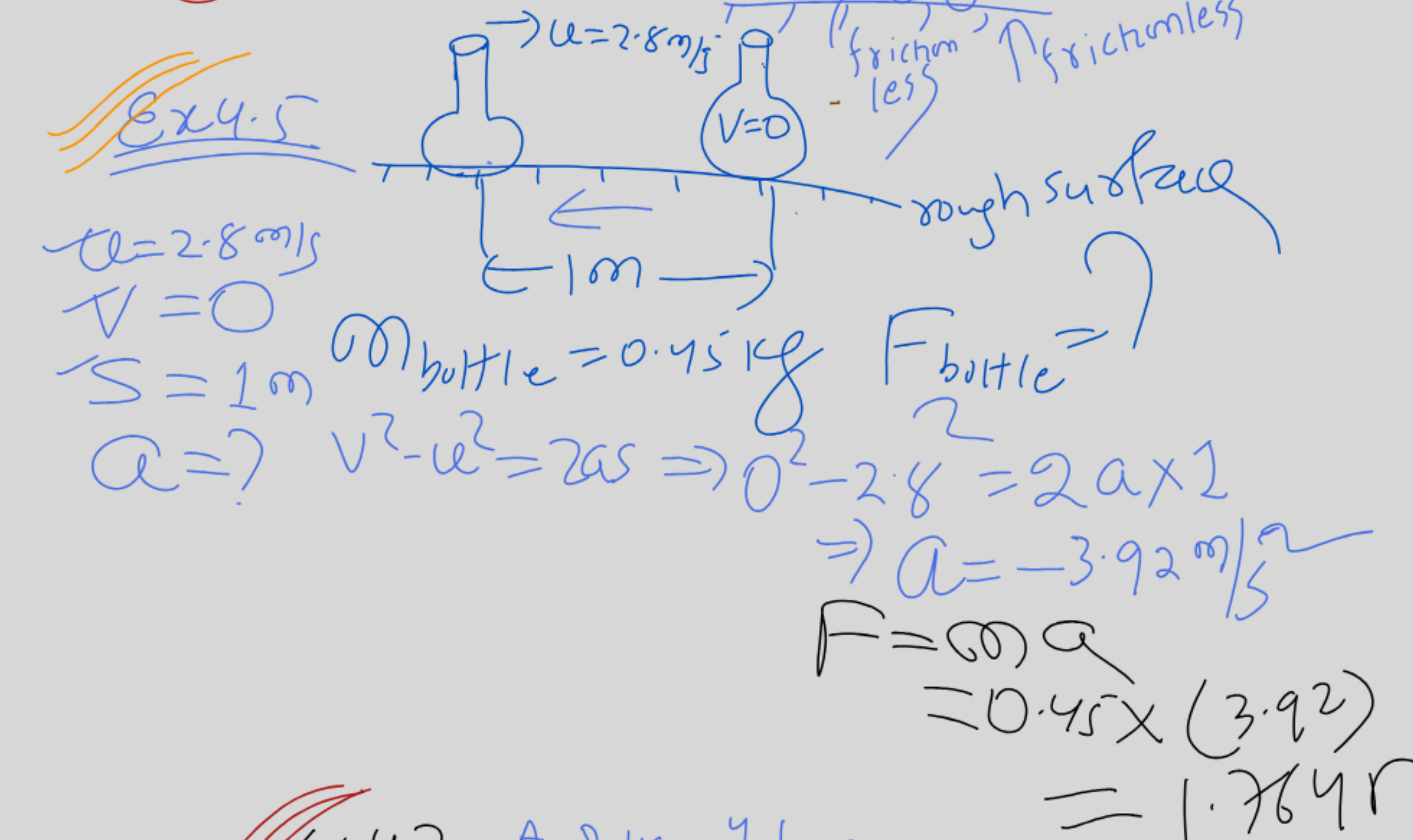
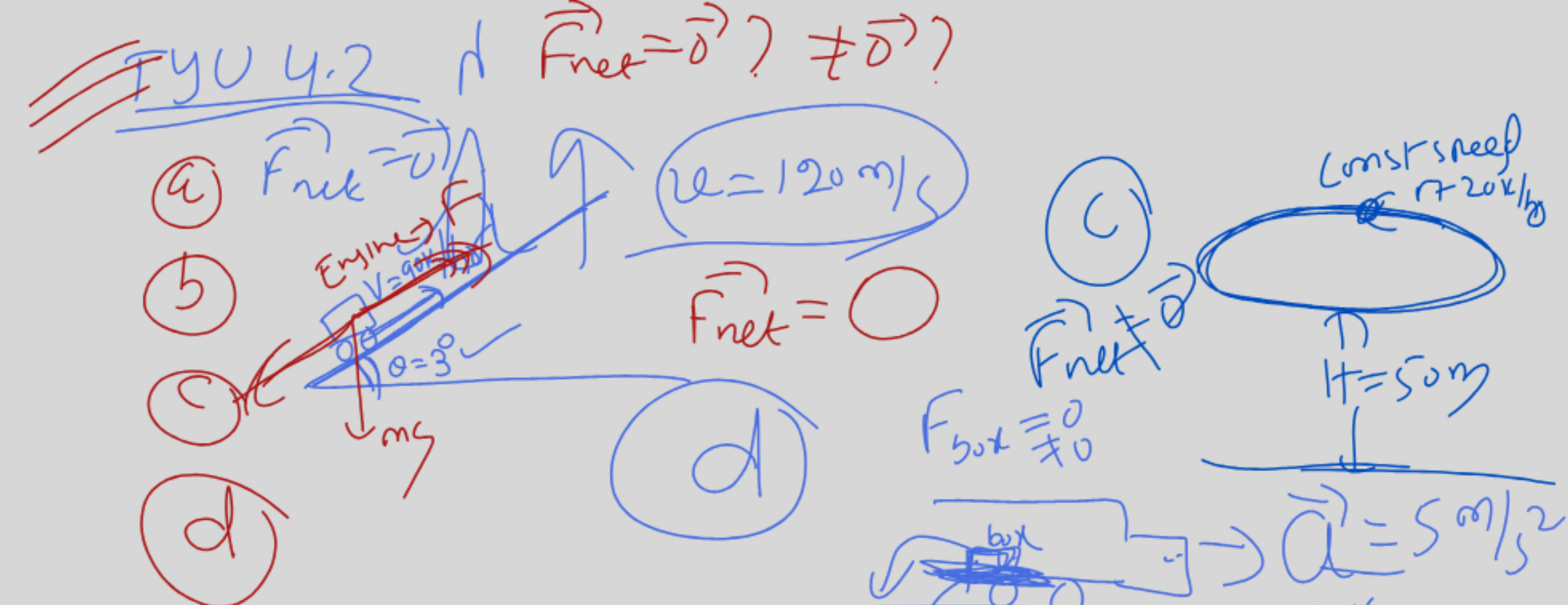
$$F_{\text{ext}} \propto \frac{d\vec{p}}{dt}$$

$$\Rightarrow \vec{F}_{\text{ext}} = K \frac{d\vec{p}}{dt} \Rightarrow \vec{F}_{\text{ext}} = \frac{d\vec{p}}{dt}$$

$$\Rightarrow \vec{F}_{\text{ext}} = \frac{d(m\vec{v})}{dt}$$

$$= m \left(\frac{d\vec{v}}{dt} \right) + \vec{v} \left(\frac{dm}{dt} \right)$$

$$\Rightarrow \vec{F}_{\text{ext}} = m\vec{a}$$

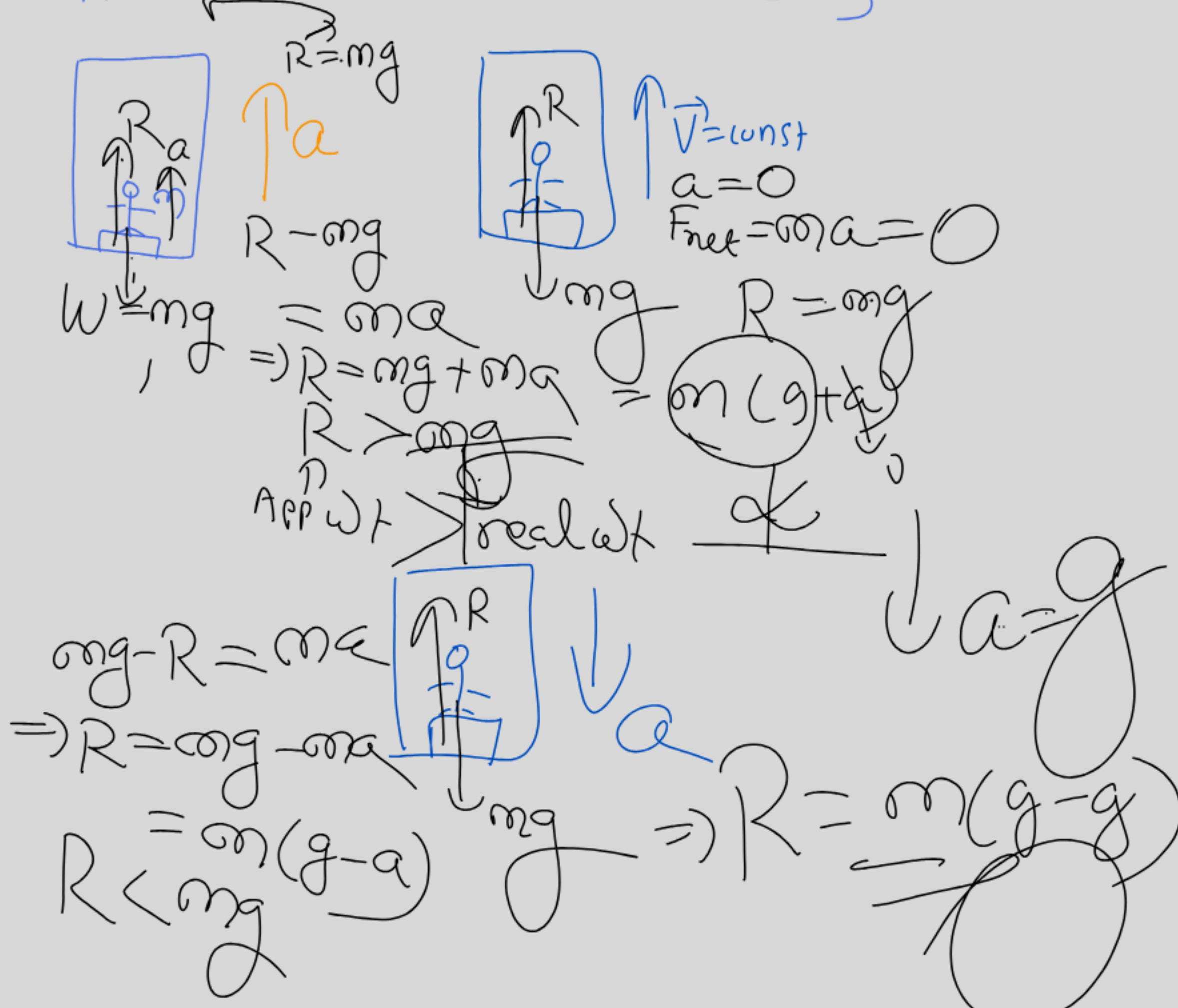


Ex 4.7 A $2.49 \times 10^4 \text{ kg}$ car is moving in the x dir & makes emergency stop.

$\vec{F}_x = -1.83 \times 10^7 \text{ N} = \vec{F}_{ext}$ $a = ?$

Sol $W = mg$
 $\Rightarrow 2.49 \times 10^4 = m \times 9.8 \Rightarrow m = 2540 \text{ kg}$
 $F = ma \Rightarrow -1.83 \times 10^7 = 2540 \times a$
 $a = -7.2 \text{ m/s}^2$

Real wt & Apparent wt:



Quiz Test 1

on 12th October (Monday)

Position

Ch. 2, 3, Ch. 4 (4.1, 4.2)

Newton's Law of Cooling.

$T(t)$ → temperature of an object at time t

T_s → temperature of the surrounding

$$\frac{dT}{dt} \propto (T - T_s) \Rightarrow \frac{dT}{dt} = k(T - T_s), T(0)$$

$$\text{Let } \underbrace{T(t) - T_s}_{y(t)} = y(t) \Rightarrow \frac{dT}{dt} = \frac{dy}{dt}$$

$$\frac{dy}{dt} = ky,$$

$$y(0) = T(0) - T_s$$

Exercise 3.8.

Q. 13. A roast turkey is taken from an oven when its temperature has reached 85°C and is placed on a room whose temperature is 22°C .

$$\frac{dT}{dt} = k(T - 22) \quad T_s = 22 \quad T(0) = 85$$

$$\text{Let } y = T - 22 \Rightarrow y(0) = T(0) - 22 = 85 - 22 = 63$$

$$\frac{dy}{dt} = ky, \quad y(0) = 63$$

$$y(t) = y(0)e^{kt} = 63e^{kt}$$

(a) If the temperature of the turkey is 65°C after half an hour, what is the temperature after 45 minutes.

$$T(30) = 65 \Rightarrow y(30) = T(30) - 22 = 65 - 22 = 43$$

$$y(30) = 43 = 63e^{30k} \Rightarrow e^{30k} = \frac{43}{63} \Rightarrow 30k = \ln\left(\frac{43}{63}\right) \Rightarrow k = \frac{1}{30} \ln\left(\frac{43}{63}\right)$$

$$y(t) = 63e^{\frac{t}{30} \ln\left(\frac{43}{63}\right)} = 63e^{\frac{\ln\left(\frac{43}{63}\right)}{30} t} = 63 \cdot \left(\frac{43}{63}\right)^{t/30}$$

$$y(45) = 63 \cdot \left(\frac{43}{63}\right)^{45/30}$$

$$T(45) = y(45) + 22 = 63 \cdot \left(\frac{43}{63}\right)^{45/30} + 22 \approx 57.52^\circ\text{C}$$

(b) when will the turkey have cooled to 40°C ?

$$T(t) = 40 \Rightarrow y(t) = T(t) - 22 = 40 - 22 = 18$$

$$18 = 63 \cdot \left(\frac{43}{63}\right)^{t/30} \Rightarrow \left(\frac{43}{63}\right)^{t/30} = \frac{18}{63}$$

Ans Q. 14

$$\Rightarrow \frac{t}{30} = \frac{\ln\left(\frac{18}{63}\right)}{\ln\left(\frac{43}{63}\right)} \Rightarrow t = 30 \left[\frac{\ln\left(\frac{18}{63}\right)}{\ln\left(\frac{43}{63}\right)} \right] \approx 98.4 \text{ minutes}$$

Q. 15. When a cold drink is taken from a refrigerator its temperature is 5°C . After 25 minutes in a 20°C room its temperature has increased to 10°C .

(a) What is the temperature of the drink after 50 minutes?

(b) When will the temperature be 15°C ?

⑤ Bitwise left shift ($\ll$)

$a \ll n \rightarrow$ shift a by 'n' bits.

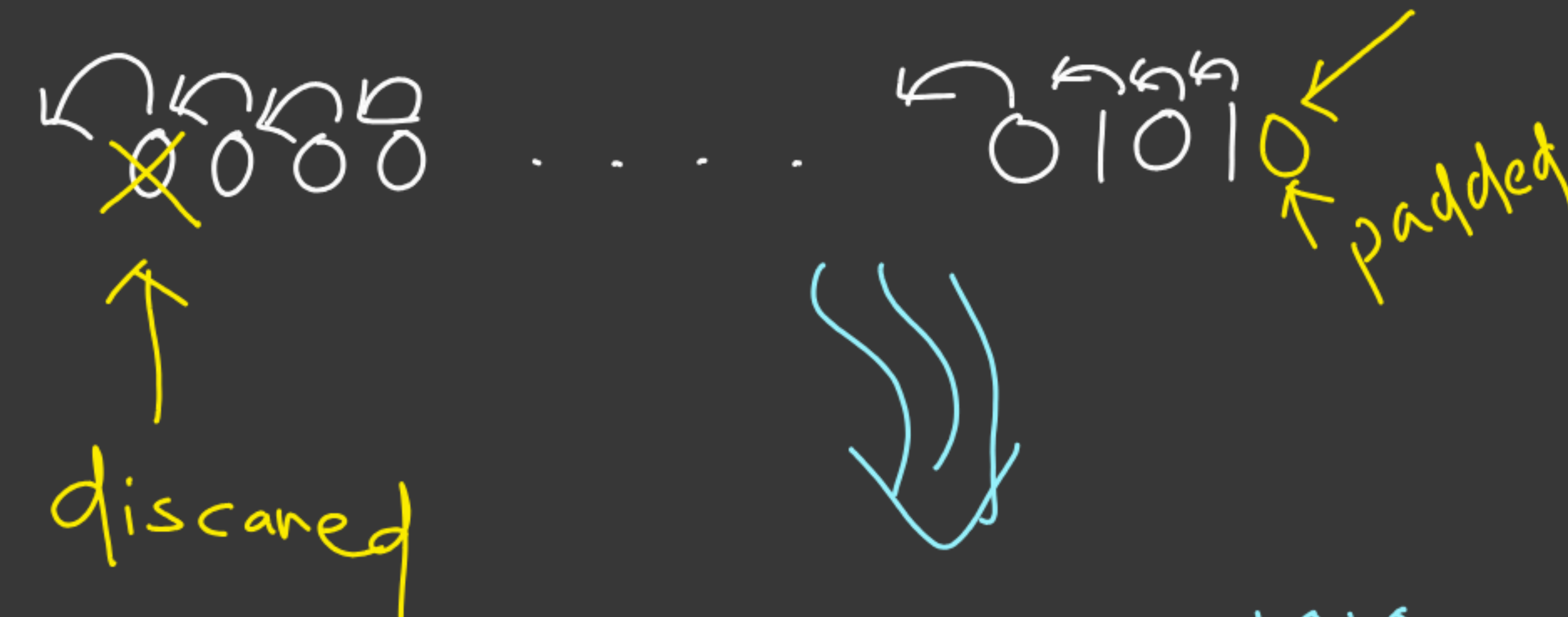
$5 \ll 1$

int a=5;

$a \ll 1$

a=5 $\Rightarrow$

$a \ll 1$



$\ll \rightarrow$ n no. of bit from RHS
 ll be discarded & 'n' no.
 of 0's in RHS ll be padded

int a=5;

int b = a << 2;

S.o.pln(b) $\rightarrow$ 20

$a \ll 2$

int b = a << 3;

S.o.pln(b) $\rightarrow$ 40

int a=5;
 int b = a << 1;
 S.o.p(b);



$5 \ll 1 \Rightarrow 10$
 $5 \ll 2 \Rightarrow 20$
 $5 \ll 3 \Rightarrow 40$
 $5 \ll 4 \Rightarrow 80$

$5 \ll 32$

no. of shift
 $a * 2$
 $5 * 2^1 = 10$
 $5 * 2^2 = 20$
 $5 * 2^3 = 40$
 $5 * 2^4 = 80$

int a = -5;

int b = a << 1;

s.o.pln(b); -10

-2147483648 to +2147483647

a = 5 $\Rightarrow$ 0000 0101

1111 1010
+ 1

-5 = 1111 1011

$2^{30} \Rightarrow 1073741824$

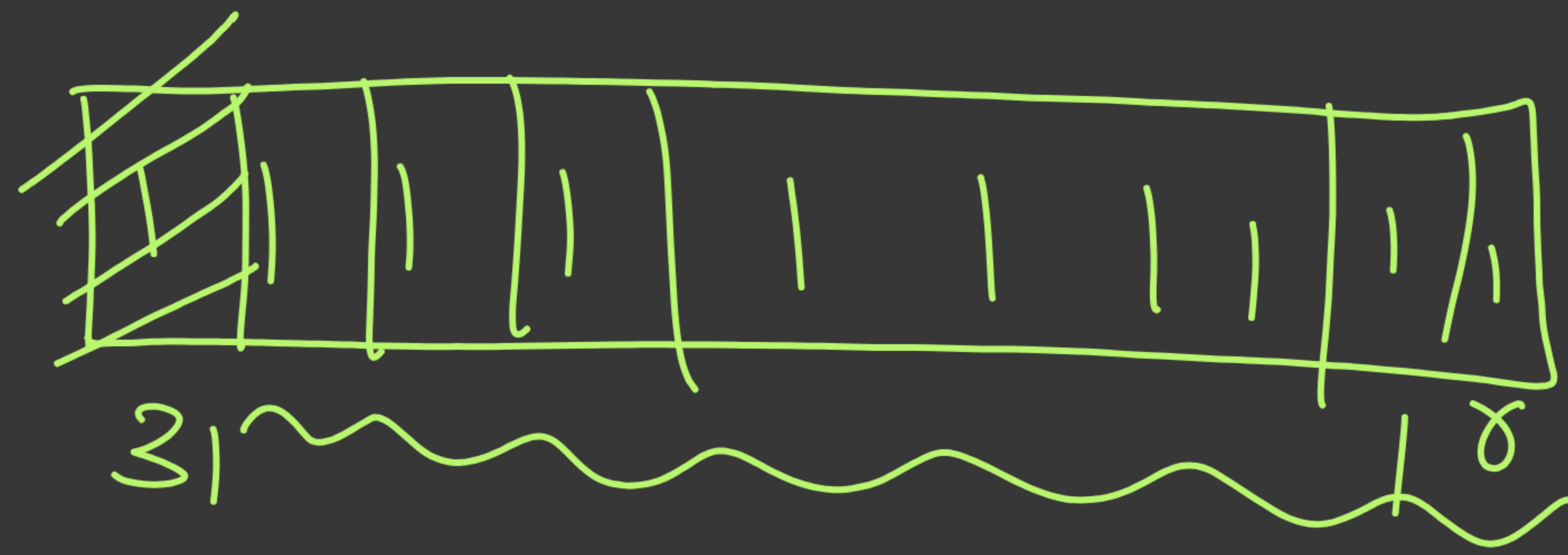
$2^{31} = 2147483648$

-10

-5 $\Rightarrow$ 1111 10110

1111 10110
000 01001 $\leftarrow$ is
+ 1

1 000 01010

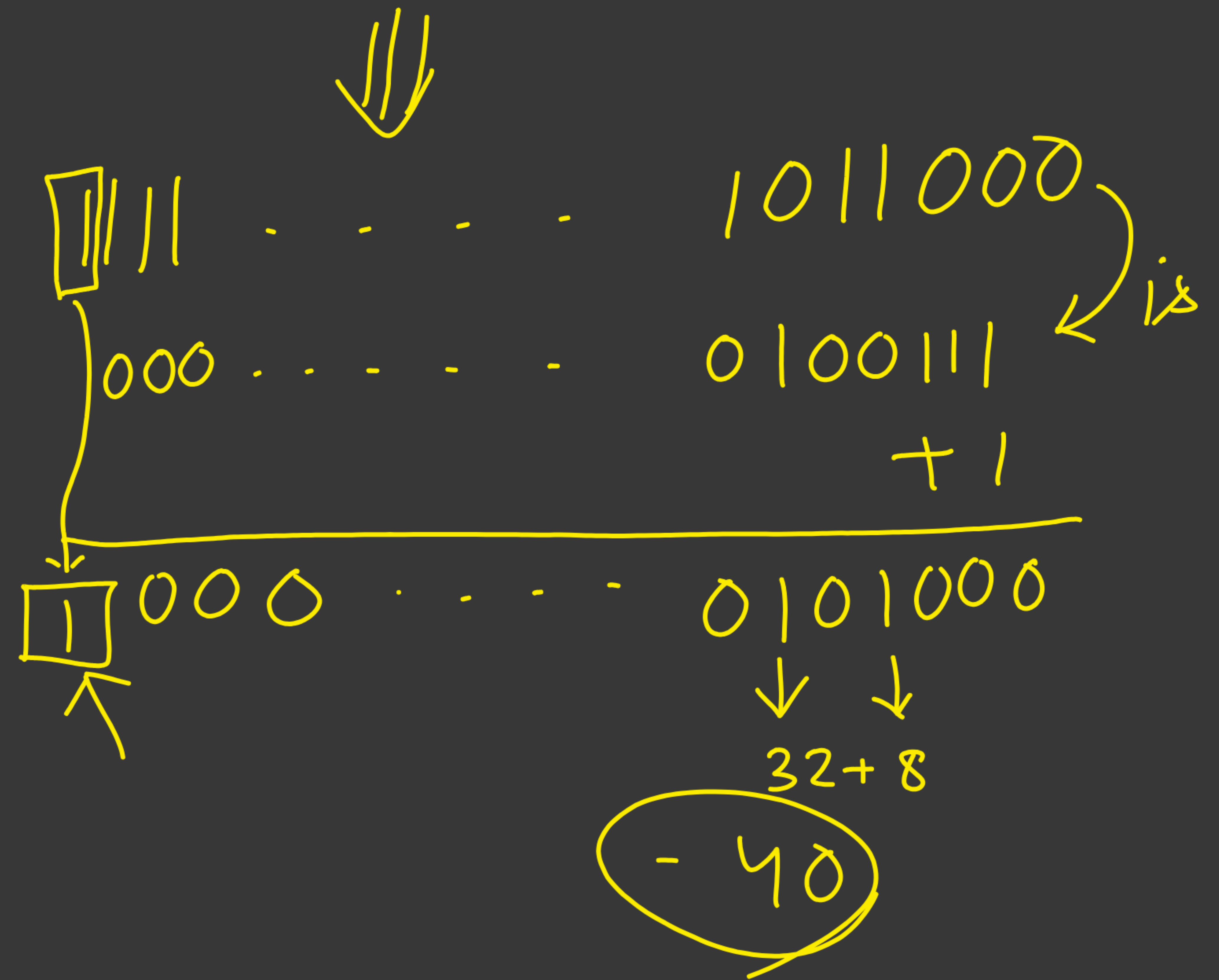


int a = -5;

int b = a << 3;

sizeof(b).

-5 $\Rightarrow$ ~~1111~~ 1011 ~~000~~



>> (Right shift)

$a \gg n \Rightarrow$ shift a by n bits

int $a=5$;

int $b = a \gg 1$;

s.o.pln(b);

RHS $\rightarrow$ bits ll be discarded

LHS $\rightarrow$ 0's ll be padded.

$$\frac{5}{2} = 2$$

$a \gg 1 \Rightarrow$

$$\frac{a}{\text{no. of shif}}$$

$$\frac{5}{2}$$

$a=5 \Rightarrow$ 00000...0101

↓↓↓

0000...0010

↓

2

$a=5 \Rightarrow$ 000000...0101

0000...0001

↓↓

1

int $a=5$;

int $b = a \gg 2$;

s.o.pln(b);

$$\frac{5}{2^2}$$

1

int $b = a \gg 3$

$$\frac{5}{2^3}$$

↓↓↓

000000...0101

0000...0000 = 0

0

int a = -5; $\ggg$
 int b = a >> 1;
 s.o.pln(b);

int b = a >> 2;
 s.o.pln(b);

int b = a >> 3;
 s.o.pln(b);

$\gg$ → Signed Right Shift

$\ggg$ → Unsigned R.S.

-5 $\Rightarrow$ ~~1~~1111 ... 1011

1111 ... 1101
 000 ... 0010
 +1
 1000 ... 0011

RHS → No. of bit
 will be discarded

LHS → No. of 1's
 will be padded

-3

-ve no.
 Right shift
 ($\gg$)

a = 5 $\Rightarrow$ ~~11111111~~ ... 1011

1111 ... 1111
 0000 ... 0000
 +1
 1000 ... 0001

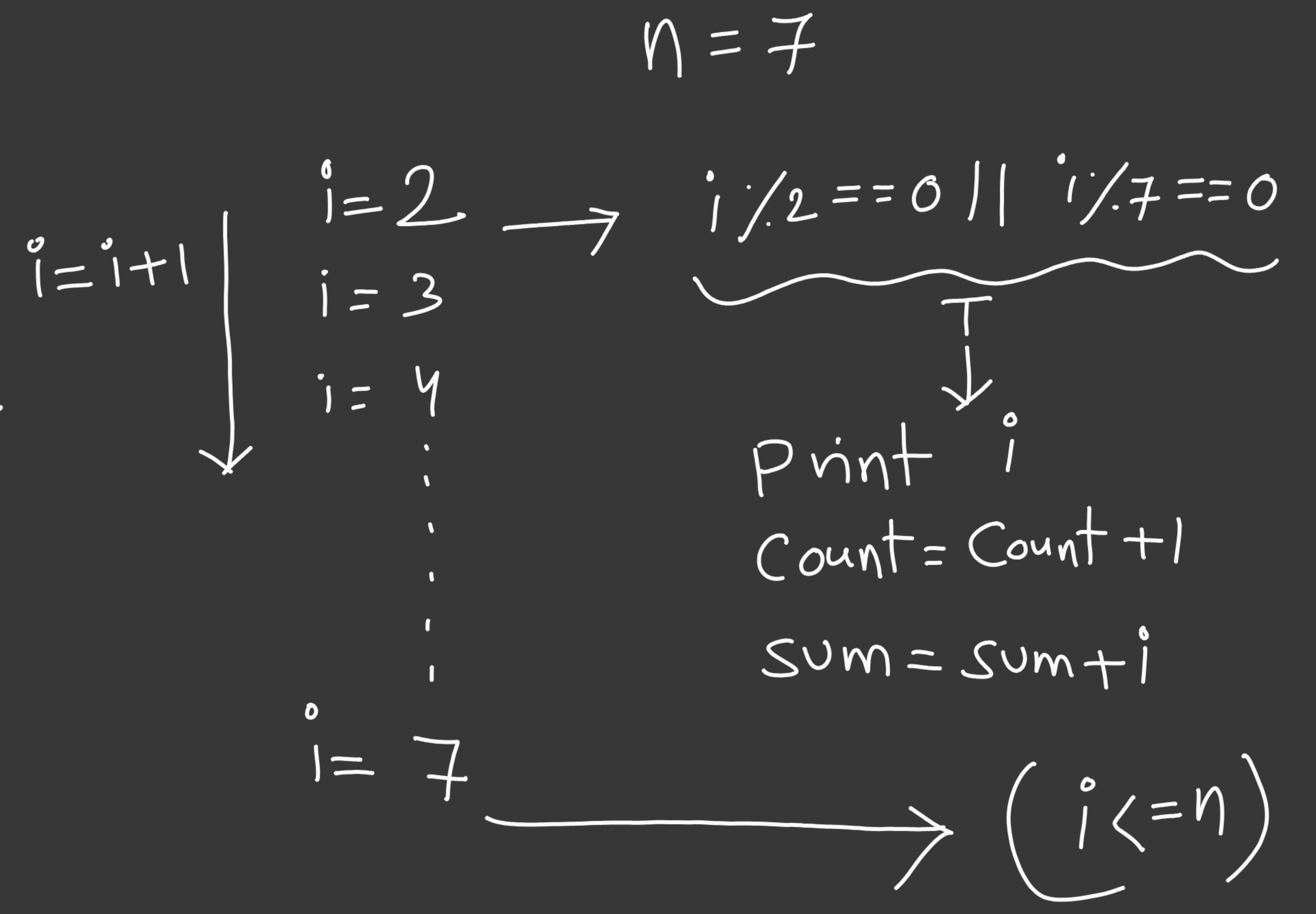
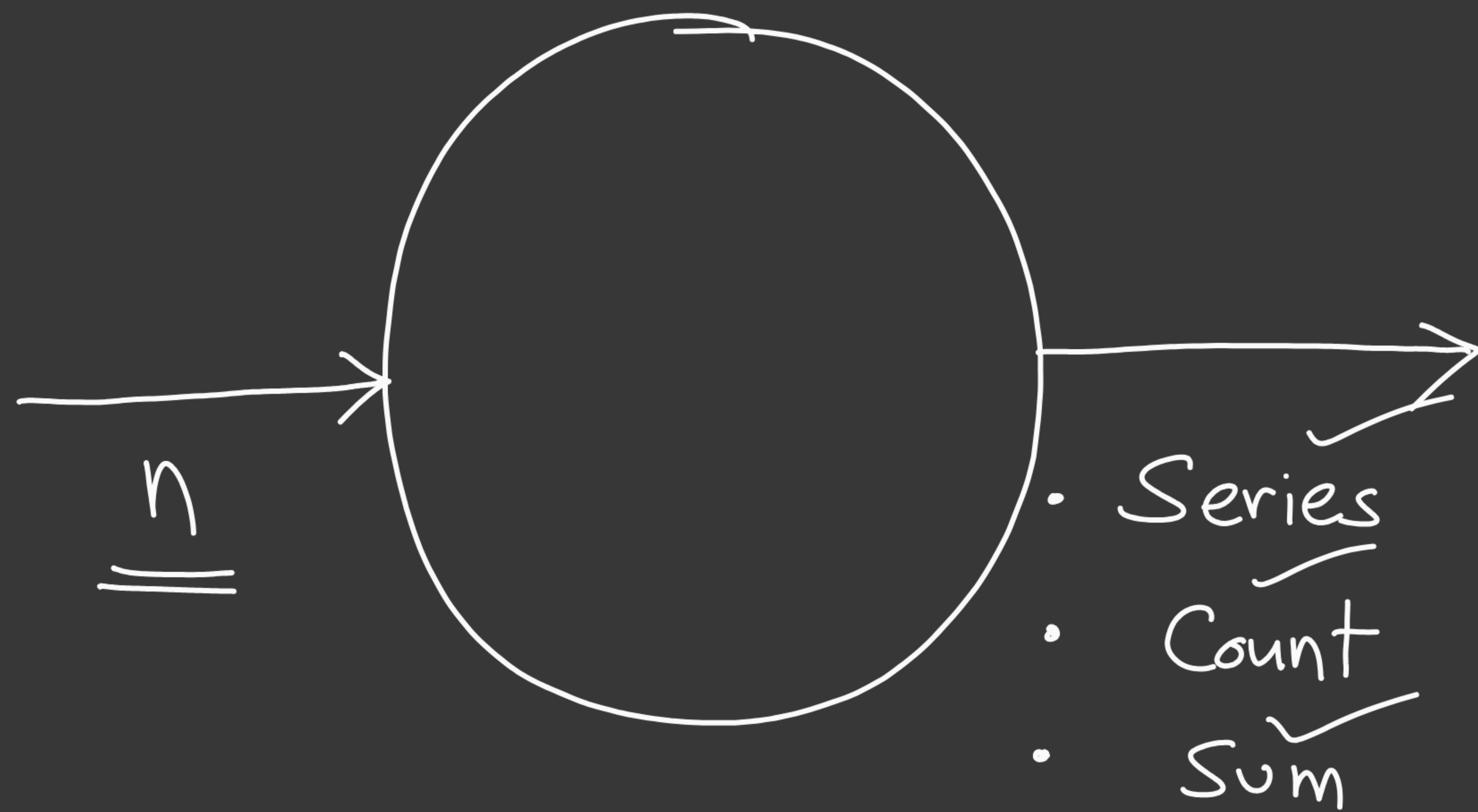
-1

Q

Q:- Print a statistics.

For a given number 'n' print the series from 2 to n which are divisible by 2 or 7.
Also Count it and perform the sum.

<u>n</u>	<u>% 2</u>	<u>% 7</u>	<u>Count</u>	<u>Sum</u>
14	(2, 4, 6, 8, 10, 12, 14)	(7, 14)	8	2 + 4 + 6 + 8 + 7 10 + 12 + 14 63
9	(2, 4, 6, 8)	(7)	5	2 + 4 + 6 + 8 + 7 = 27



Algorithm

Step 1:- Input n

Step 2:- Initialize $i=2$, $sum=0$, $Count=0$

Step 3:- Repeat step 4 to 8 until $i \leq n$

Step 4:- if $(i \% 2 == 0 \parallel i \% 7 == 0)$

Step 5:- Print i ✓

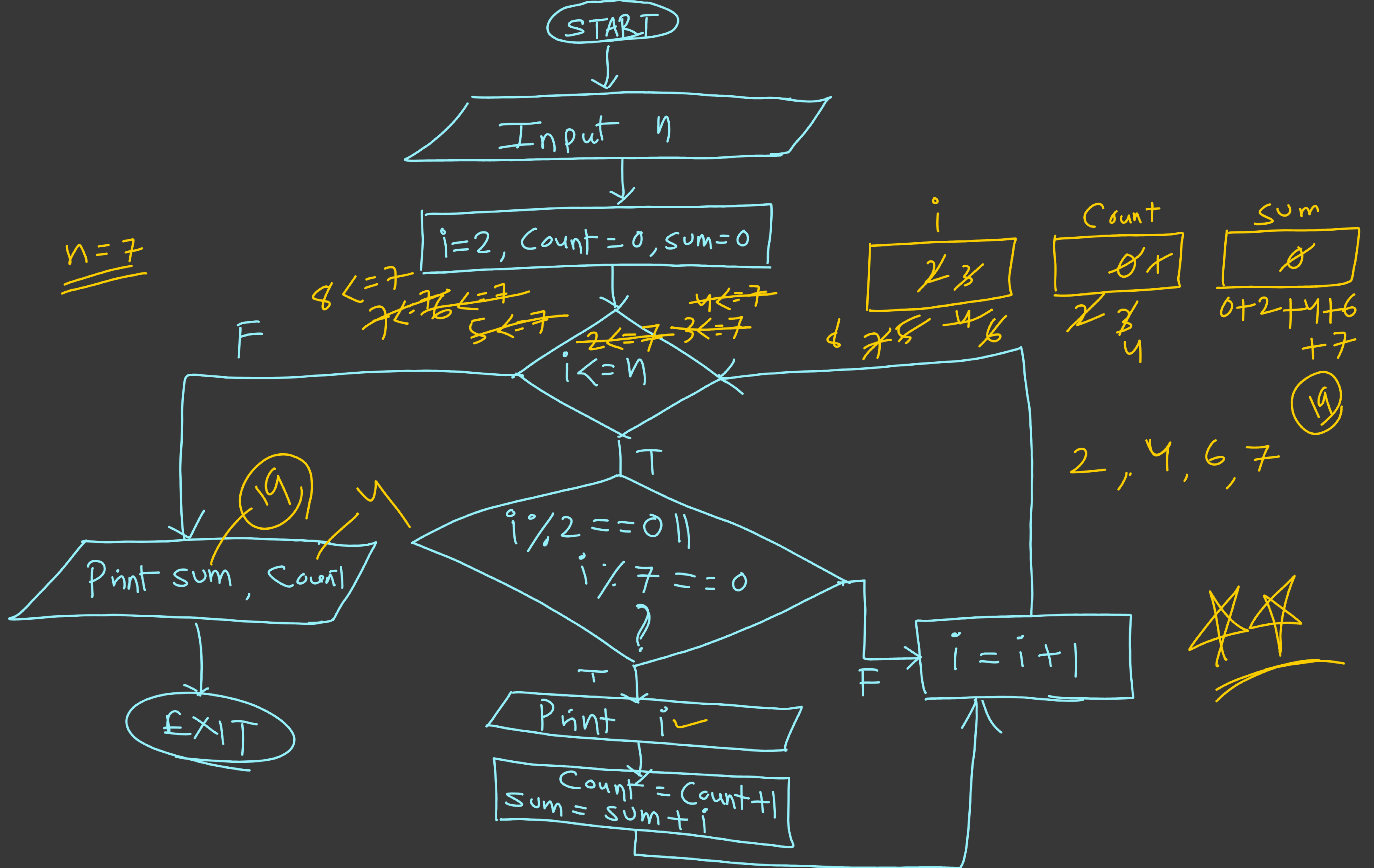
Step 6:- $Count = Count + 1$

Step 7:- $sum = sum + i$

Step 8:- $i = i + 1$

Step 9:- Print $Count$ ✓, Print sum ✓

Step 10:- Exit.



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int a = Integer.MAXVALUE;
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Rang

$$\begin{array}{rcl} -2147483648 & \text{to} & +2147483647 \\ (\text{MIN}) & & (\text{MAX}) \end{array}$$

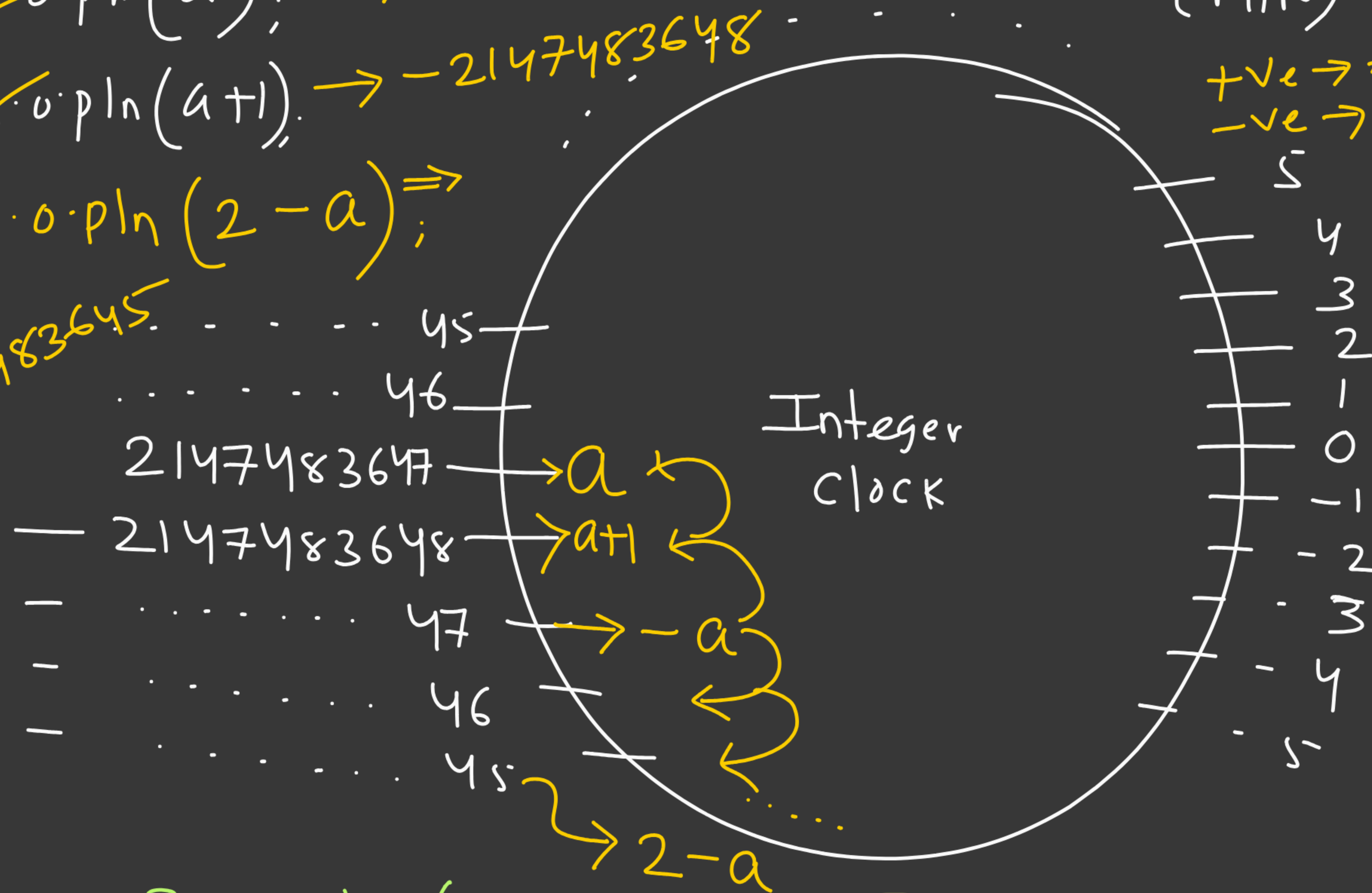
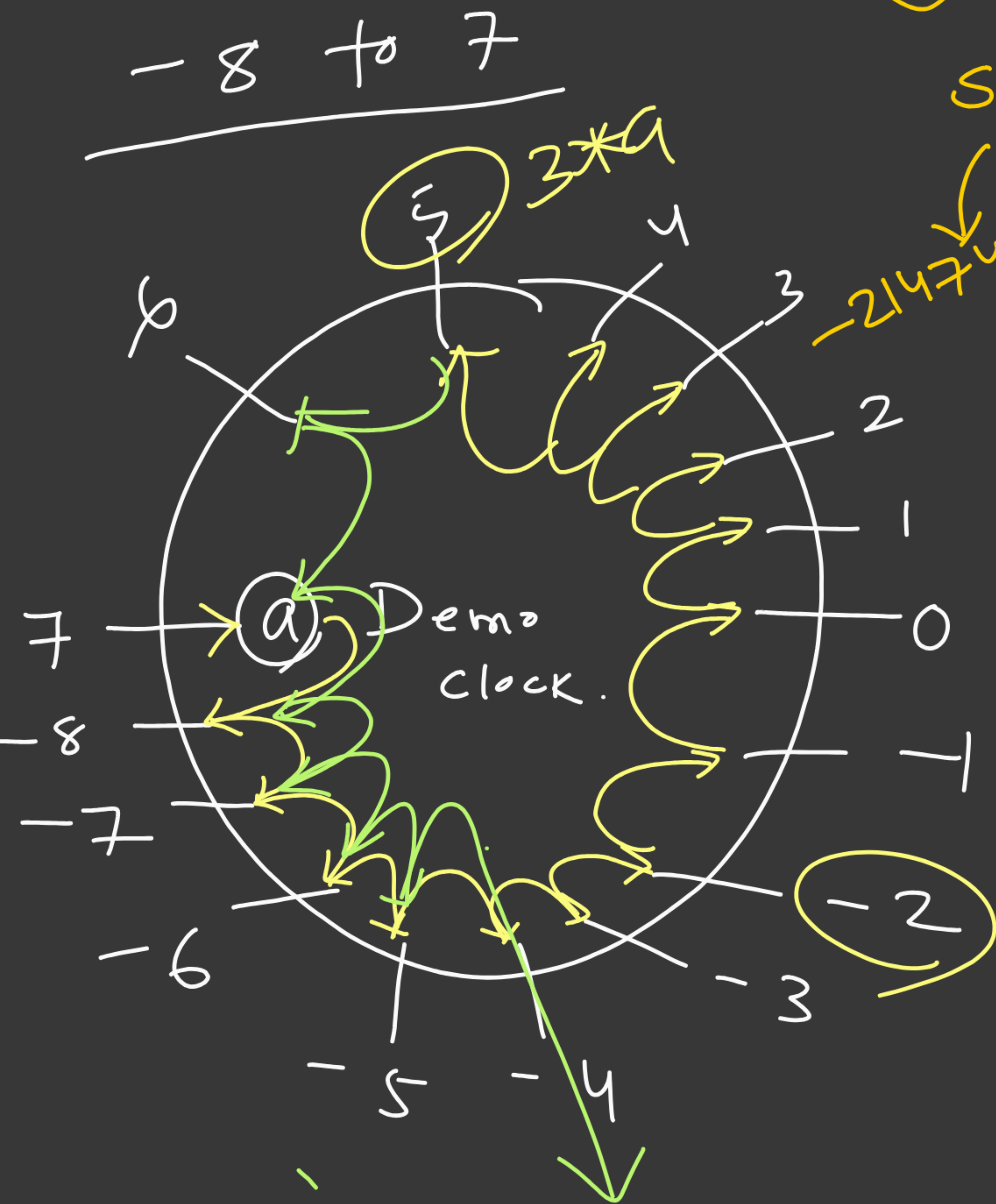
Soln (a): $\rightarrow 2147483647$

$$\text{S.O.P.} \ln(a+1) \rightarrow -2147483648$$
$$S.O.Pln(2-a) \Rightarrow$$

+ve $\rightarrow$ Anticlockwise
-ve $\rightarrow$ " "

$$s''_{\text{opt}}(-2-a);$$

21474 83647

$$S.O.p.l_n(2 * a)$$

$$S.O. \ln(2 * a) = -2$$
$$s.o.pln(3*a) \Rightarrow 2147483645$$
$$S \cdot \phi \ln(y * a) - y$$